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26 August 1955
SUPERSEDING
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1 October 1943

MILITARY SPECIFICATION

PACKING, PREFORMED, CARBON, AND CARBON STOCK, PACKING

All interested bureaus of the Department of the Navy have concurred in the use of this specification.

1 SCOPE

1 1 Scope - This specification covers carbon packing material (1700) for rotary rod and shaft service

1 2 Classification - The packing shall be of the following classes, as specified (see 6 2).

Class 1 - Soft
Class 2 - Medium
Class 3 - Hard

2 APPLICABLE DOCUMENTS

2 1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification

SPECIFICATIONS

FEDERAL
NN-B-591 - Boxes, Fiberboard, Wood-Cleated (for Domestic Shipment).
NN-B-621 - Boxes, Wood, Wirebound (for Domestic Shipment)
LLL-B-631 - Boxes, Fiber, Corrugated (for Domestic Shipment)
LLL-B-636 - Boxes, Fiber, Solid (for Domestic Shipment)
PPP-B-[illegible]75 - Boxes, Straparound
PPP-B-601 - Boxes, Wood, Cleated Plywood

MILITARY
JAN-P-103 - Packaging and Packing for Overseas Shipment - Boxes, Wood Cleated, Solid Fiberboard.
JAN-P-106 - Packaging and Packing for Overseas Shipment - Boxes, Wood Nailed
JAN-P-108 - Packaging and Packing for Overseas Shipment - Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior
MIL-P-116 - Preservation, Methods of
JAN-W-[illegible]62 - Wire, Nickel-Alloy, Spring (Heat-Resistant and Age-Hardenable)
MIL-B-4299 - Boxes, Paperboard, Metal-Stayed
MIL-N-7786 - Nickel-Chromium Alloy, Sheet and Strip, Age-Hardenable Annealed
MIL-B-10377 - Boxes, Wood-Cleated, Veneer, Paper Overlaid
MIL-L-10547 - Liners, Case, Waterproof

FED. SUP CLASS
5330

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NAVY DEPARTMENT
General Specifications for Inspection of Material

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer)

2 2 Other publications - The following document forms a part of this specification. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply

CONSOLIDATED CLASSIFICATION COMMITTEE
Consolidated Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3 REQUIREMENTS

3 1 Material - Packing material shall be composed of refined coke, lubricating graphite, or refined coke and lubricating graphite, and a suitable binder, and shall contain no abrasive or other material which may affect its serviceability

3 1 1 Composition - Packing material shall conform to the composition shown in table I

Table I - Composition

Class	Field carbon minimum	Ash maximum	Sulfur [1]
	Percent	Percent	Percent
1	96.0	1 5	1
2	94 0	2.0	1
3	93 0	3.0	1

[1] Sulfur conten shall be determined as a matter of record

3 1 2 Character of ash - When the packing material is tested as specified in 4 3 2, the ash shall be soft and fluffy and shall contain no hard or abrasive particles.

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3.2 Physical requirements. - The packing material shall conform to the physical requirements shown in table II.

Table II - Physical requirements.

Class	Hardness, numerical, scleroscope (Shore)	Apparent density (minimum) pounds per cubic inch	Porosity (maximum) percent	Abrasion loss (maximum) milligrams per revolution
1	20 - 50	-----	---	---
2	30 - 60	0.0542	18.0	2.0
3	60 - 90	-----	---	---

3.3 Form and dimensions - The packing material shall be furnished in solid carbon stock or ring form and shall be of the dimensions specified (see 6.2)

3.3.1 Rings. -

3.3.1.1 The joints of the rings shall be of the butt type.

3.3.1.2 The flat sides of each ring shall be parallel and shall have a smooth, even surface. The inside diameter shall be ground true and a perfect right angles to the side of the ring.

3.3.1.3 Unless otherwise specified in the contract or order, the inside diameter of all rings shall be finished to the nominal shaft diameter with a permissible variation of minus nothing and plus 0.001 inch per inch of nominal shaft diameter. The diameter shall be measured with segments concentric and all joints butted.

3.3.1.4 The segments of each ring shall be marked so that they may be assembled in the same relative position as when manufactured.

3.3.1.5 Each ring shall be furnished with one coiled garter spring made of either commercial cold-drawn, spring-temper, corrosion-resisting steel wire or commercial nickel-copper-aluminum wire. On sizes for shafting larger than 4 inches in diameter, a corrugated spring of commercial tungsten-chrome-vanadium steel, or a corrosion-resisting steel band whose ends are held together with a tungsten-chrome-vanadium coil spring, may be furnished. The strength of the springs shall be such that an elongation of not more than 15 percent shall be sufficient to hold the ring segments lightly against the shaft. For steam turbines, springs shall be of one of the following materials:

(a) JAN-W-562, class 2 (spring temper), or
(b) JAN-W-562, class 2 (spring temper), except for chemical composition as follows

Element	Percent
Nickel (including cobalt)	70.00 min
Chromium	14.00 - 17.00
Titanium	2.00 - 2.75
Aluminum	0.40 - 1.00
Iron	5.00 - 9.00
Manganese	1.00 max.
Silicon	0.70 max.
Copper	0.50 max.
Carbon	0.08 max
Sulfur	0.08 max

(c) MIL-N-7786, condition and finish as specified (see 6.2)

Note - Items (a) and (c) above, to the compositions shown in Specifications JAN-W-562 and MIL-N-7786, respectively are known commercially as "Inconel X", and item (b) as "Inconel W".

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3 4 Workmanship - Packing material shall be free from laminations, shrinkage, cracks, or other imperfections

4 QUALITY ASSURANCE PROVISIONS

4.1 Sampling -

4.1.1 Lot - For purposes of sampling, a lot shall consist of all ring assemblies or all solid blanks of one class and size produced in one plant under essentially the same conditions and offered for delivery at one time

4 1 2 Sampling for visual and dimensional inspection (at the place of manufacture) - A random sample of ring assemblies or solid carbon stock shall be selected by the Government inspector in accordance with table III from each inspection lot of material offered for Government inspection of visual and dimensional characteristics with lot acceptance based on the sampling inspection requirements in accordance with Standard MIL-STD-105

Table III - Sampling for visual and dimensional inspection
AQL (approx.) = 2.5 percent defective.

Number of ring assemblies or solid carbon stock in inspection lot	Number of ring assemblies or solid carbon stock in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	2	3
501 to 800	50	3	4
801 to 1,300	75	4	5
1,301 and over	110	6	7

4 1 3 Sampling for lot acceptance tests (at the place of manufacture) -

4 1 3 1 Sampling of ring assemblies - The Government inspector shall select at random from each lot one ring assembly and one carbon stock, at least 9 inches by 9 inches, of the same composition as that of the rings for the tests specified in 4 2.2 1

4 1 3 2 Sampling of carbon stock - The Government inspector shall select at random from each lot one carbon stock for the tests specified in 4 2 2 2

4.1 4 Sampling for production check tests (at a Government laboratory) -

4 1 4 1 Sampling of ring assemblies - From the first lot of ring assemblies of each class and size offered for delivery under a contract or order, and thereafter from one lot in each group of 10 successive lots of ring assemblies of the same class and size, which have passed inspection at the place of manufacture, the Government inspector shall select one complete ring and spring assembly and one carbon stock at least 9 inches by 9 inches, of the same composition as that of the rings Unless

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otherwise specified in the contract or order, the samples so selected shall be forwarded by the Government inspector to the U.S Naval Engineering Experiment Station, Annapolis, Maryland, for the tests specified in 4.2 3

4.1.4.2 Sampling for carbon stock - From the first lot of carbon stock of each class and size offered for delivery under a contract or order, and thereafter from one lot in each group of 10 successive lots of carbon stock of the same class and size which have passed inspection at the place of manufacture, the Government inspector shall select one carbon stock. Unless otherwise specified in the contract or order, the samples so selected shall be forwarded by the Government inspector to the U.S. Naval Engineering Experiment Station, Annapolis, Maryland, for the tests specified in 4.2.3

4 2 Inspection and tests -

4.2.1 Visual and dimensional inspection (at the place of manufacture). -

4 2 1.1 Ring assemblies - Each of the sample ring assemblies selected in accordance with 4 1 2 shall be visually inspected by the Government inspector to verify compliance with this specification.

4 2 1 1 1 Dimensions - The dimensions of each sample ring assembly selected in accordance with 4 1 2 shall be determined by the method specified in 4.3.4.

4 2 1 2 Carbon stock - Each of the sample carbon stocks selected in accordance with 4 1 2 shall be visually and dimensionally inspected by the Government inspector to verify compliance with this specification

4 2.1.3 Action in case of failure - Any ring assembly or carbon stock containing one or more visual or dimensional defects shall be rejected, and if the number of defective ring assemblies or carbon stocks in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for Government inspection provided that the contractor has removed or repaired all nonconforming ring assemblies or carbon stocks

4 2 2 Lot acceptance tests (at the place of manufacture) -

4 2 2 1 Lot acceptance tests of ring assemblies - The sample ring assembly selected in accordance with 4 1 3 1 shall be subjected to the test specified in 4 3 3 1 and the sample carbon stock shall be subjected to the tests specified in 4 3.3.1 and 4 3 3 2. If any one of the samples tested is found to be not in conformance with this specification, the lot which it represents shall be rejected. Rejected lots may be resubmitted for acceptance tests provided the contractor has removed or reworked all nonconforming products.

4.2.2 2 Lot acceptance tests of carbon stocks - The sample carbon stocks selected in accordance with 4 1 3 2 shall be subjected to the tests specified in 4 3.3.1 and 4 3 3.2 If the sample tested is found to be not in accordance with this specification, the lot which it represents shall be rejected Rejected lots may be resubmitted for acceptance tests provided the contractor has removed or reworked all nonconforming products

4.2 3 Production check tests (at a Government laboratory) - The sample specimens selected in accordance with 4 1 4 shall be subjected to the tests specified in 4.3

4.2 3 1 Action in case of failure - Acceptance of the first lot offered for delivery under a contract or order shall be withheld until a satisfactory report is received on the production check test sample Thenceforth, except as hereinafter specified, acceptance and rejection of lots shall normally be on the basis of the sampling, inspection, and tests specified in 4 1.2, 4.1 3, 4.2.1 and 4 2 2, and

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acceptance shall not be withheld pending receipt of production check test reports unless, in the opinion of the Government inspector, circumstances require such action However, upon receipt of an unsatisfactory test report on a production check test sample, the Government inspector shall select additional samples from every subsequent lot offered for delivery, The sample so selected shall be submitted to the U.S. Naval Engineering Experiment Station, Annapolis, Maryland, and shall there be subjected to the test or tests wherein failure was observed Lots shall then be accepted only upon receipt of a satisfactory test report on the samples so selected Additional testing shall be discontinued and lot acceptance returned to the normal basis when four successive lots have been accepted

4 3 Test procedures - The tests shall be performed as follows:

4 3 1 Chemical analysis -

4 3 1 1 Fixed carbon - Fixed carbon shall be computed as follows: 100 minus (moisture plus ash plus volatile matter) equals percentage of fixed carbon

4 3 1 2 Moisture - The oven shall be so constructed as to have uniform temperature in all parts and a minimum of air space Provision shall be made for renewing the air in the oven at the rate of two to four times a minute, with the air dried by passing it through concentrated H_2SO_4 Heat an empty capsule or shallow dish of convenient size to 105° Centigrade (C) (221° Fahrenheit (F.)), stopper or cover, cool over concentrated H_2SO_4 (specific gravity 1.84) for 30 minutes, and weigh Dip out, with a spoon or spatula, 5 grams (gm) of the sample, which has been ground to pass a No 20 sieve, heat 1-1½ hours, cool, and weigh The difference between the original and the final weight is moisture Compute to a percentage basis

4 3 1 3 Ash - A gas or electric muffle furnace having a good air circulation and capable of having its temperature regulated between 700° and 750°C (1292° and 1382°F.) shall be used Place a porcelain capsule or shallow dish of convenient size, containing dried material from the moisture determination, in a cold muffle furnace or on the hearth at low temperature, and gradually heat to redness at such a rate as to avoid mechanical loss from too rapid expulsion of volatile matter Finish the ignition to constant weight at a temperature between 700° and 750°C. (1292° and 1382°F.) Cool in a desiccator and weigh as soon as cold

4 3 1 4 Volatile matter - Use a platinum crucible with tightly fitting cover The crucible should be of not less than 10 nor more than 20 cubic centimeters (cc) capacity, not less than 25 nor more than 35 millimeters (m.m.) in diameter, not less than 30 nor more than 35 m.m. in height Use a verticle electric tube furnace, or a gas or electrically heated muffle furnace regulated to maintain a temperature of 950°C ± 20°C (1742°F ± 36°F) in the crucible, as shown by a thermocouple kept in the furnace Weigh 1 gm of the packing ground to a fineness to pass a No 20 seive in a weighed 10 to 20 cc platinum crucible, close with capsule cover, and place on platinum or nichrome wire supports in the furnace chamber, which shall be at a temperature of 950°C. ± 20°C (1742°F. ± 36°F) After the more rapid discharge of volatile matter has subsided as shown by the disappearance of the luminous flame, tap the cover lightly to more perfectly seal the crucible and thus guard against the admission of air After heating exactly 7 minutes, remove the crucible from the furnace and, without disturbing the cover, allow it to cool Weigh as soon as cold. The loss of weight minus moisture equals the volatile matter

4 3 1.5 Sulfur - The sulfur content shall be determined by the Parr sodium peroxide bomb method Place 1 gm of $KClO_4$ in the bomb and add a 0.5 gm sample, and mix thoroughly with a glass rod Then add approximately 15 gm of sulfur-free Na_2O_2, close the bomb, and mix thoroughly by shaking Fasten the cover securely to the bomb and ignite the charge by applying a sharply pointed flame from a blast lamp to the bottom of the bomb for a brief period, or by electric ignition, according to the type of bomb used Place bomb inside a piece of steel pipe when charge is ignited to prevent possible injury to the operator if the bomb should burst Allow 1 minute for complete combustion to

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take place after ignition, then cool under a stream of water from tap or in a vessel of water Remove the cover from bomb, place bomb on its side in a 400 milliliter (mℓ) beaker, and wash of the cover with a fine jet of hot distilled water Place a watch glass over the beaker and cautiously add approximately 100 mℓ of hot distilled water After the contents of the beaker have dissolved, remove and rinse it carefully with distilled water Add, slowly, concentrated HCℓ to the neutral point then add 1 to 2 mℓ of the acid in excess Filter through qualitative filter paper into a 600 mℓ beaker and wash thoroughly five or six times with distilled water Dilute the filtrate to approximately 400 mℓ and precipitate the sulfur with $BaC\ell_2$. A correction shall be applied by running a blank on the reagents used The percentage of sulfur shall be calculated as follows

$$\text{Percentage of sulfur} = \frac{(\text{weight of } BaSO_4 - \text{blank}) \times 13.74}{\text{weight of sample}}$$

4 3 2 Character of the ash - A sample of the packing material shall be burned to complete combustion at a temperature of 1000°C (1832°F.). Upon cooling, the ash shall be examined to determine if it contains any hard or abrasive particles

4 3 3 Physical requirements -

4 3 3 1 Hardness - The hardness of the ring or carbon stock shall be determined by means of the scleroscope (Shore) hardness instrument. Five determinations shall be made on each sample and the results recorded as the average of the five determinations

4 3 3 2 Apparent density - The apparent density shall be determined on a carbon stock, at least 3 inches by 3 inches, by dividing the weight of the material by its volume and shall be expressed in pounds per cubic inch Accuracy of the measurements shall be as follows.

Length and width	0.01 inch
Thickness	.001 inch
Weight	.05 pound.

The apparent density shall be the average of duplicate determinations calculated as follows:

$$\text{Apparent density} = \frac{\text{Weight in pounds}}{\text{Width inches} \times \text{length inches} \times \text{thickness inches}}$$

4 3 3 3 Porosity - The porosity shall be determined on three samples, each approximately 1 inch square, cut from different locations on the carbon stock The samples shall be weighed on an analytical balance. Each sample shall be placed in a small beaker in which there is a wire gauze to prevent the sample from resting on the bottom of the beaker The beaker shall be filled with sufficient distilled water to cover the sample to a depth of 1/2 inch The beaker with sample shall be placed in a vacuum desiccator and evacuated to a pressure of 20 mm. of mercury This pressure shall be maintained for 1 hour or until air ceases to rise from the sample Then the system shall be allowed to return to atmospheric pressure The sample shall remain in the water for 5 minutes before removal. When removed, the sample shall be blotted lightly with filter paper to remove excess surface water and immediately reweighed The porosity shall be the average of three determinations calculated as follows:

$$\text{Porosity, percent by weight} = \frac{\text{Final weight} - \text{initial weight}}{\text{initial weight}} \times 100$$

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4 3 3 4 Abrasion resistance -

4 3 3 4 1 Abraser machine - Abrasion resistance shall be measured on an abraser (Research Model, Taber) This instrument consists essentially of 2 abrasive wheels and a motor driven specimen table The 4-inch diameter specimen table rotates at 70 revolutions per minute and actuates an automatic revolution counter Abrasion is obtained by the action of the 2 abrasive wheels revolving in contact with the specimen. The wheels are mounted on dual, hinged heads, weighted so that each exerts a 500-gram load on the test specimen The wheels revolve in opposite directions and are positioned off-center of the specimen table, producing a radial action on the specimen The wheels are of a coarse grade (H-22 Calibrade, nonresilient silicon carbide.

4 3 3 4 2 Procedure - The abrasive wheels are refaced with emery paper before each determination and brushed free of extraneous material with a stiff but nonscratching brush A specimen, 0 18 + 0 03 inch thick by 4 inches in diameter with a 1/4 inch hole in the center, shall be cut from the carbon blank. The specimen shall be weighed on an analytical balance and then mounted on the abrading table in accordance with the operating instructions for the instrument The abrasive wheels shall be lowered into place and the motor started The test shall consist of 500 revolutions of the specimen table During the test, a vacuum pickup shall be used to remove carbon dust from the specimen. The vacuum pickup shall be placed 1/8 inch above the track left by the wheels and shall have a suction sufficient to remove all carbon and abrasive matter At the end of the test the specimen shall be brushed free of extraneous matter with a soft camels hair brush and reweighed The abrasion resistance shall be the average of duplicate determinations calculated as follows

$$\text{Abrasion resistance, milligram per revolution} = \frac{\text{Initial weight - final weight}}{500} \times 1000$$

(Weights are expressed in grams)

4 3 4 Dimensions of ring assemblies. - In checking dimensions, the segments shall be assembled about a plug or disk gage and held by means of the garter spring or rubber bands. For large rings, a flexible steel clamping band may be used. The ring shall be firmly seated on the gage by rotating same and manipulating the joints until it is properly lined about the gage The rings shall then be placed over an aperture in the inspection table which encloses a light source, and visually inspected for any emission of light between the joints Emission of light between the surfaces named will be considered sufficient cause for rejection The gages shall be smaller in diameter than the specified inside diameter of the ring, as follows

0 0005 inch for rings less than 2 inches in diameter
001 inch for rings 2 to 3 inches, inclusive, in diameter
0015 inch for rings over 3 inches in diameter

4 4 Preproduction pilot pack testing - A preproduction pilot package shall be tested to determine compliance with the rough handling test of Specification MIL-P-116 The test shall be accomplished on the unit package in lieu of the shipping container

4 5 Inspection procedures - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material

5 PREPARATION FOR DELIVERY

5 1 Packaging -

5 1.1 For domestic shipment, immediate use - Packaging shall be in accordance with the manufacturer's commercial practice

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5.1.2 For domestic shipment and storage and overseas shipment - Each complete carbon packing ring with coiled garter spring shall be adequately positioned and cushioned with die cut pads or cellulosic, rubberized, or other suitable cushioning material in accordance with method III of Specification MIL-P-116 in a metal-edge or fiberboard box conforming to Specification MIL-B-4229, LLL-B-631, or LLL-B-636 The coiled garter spring for use with the packing ring shall be preserved with preservative type P-7, P-8, or P-9 and sealed in a heat-sealed greaseproof bag in accordance with method IC-3 of Specification MIL-P-116 prior to inclusion within the unit package with the packing ring. Solid carbon stocks shall be packaged and packed in containers by method III of Specification MIL-P-116 Containers for carbon stock shall conform to 5.2.2 or 5.2.3, as applicable Quantity of carbon stock per intermediate and shipping container shall be as specified (see 6.2).

5.1.2.1 Intermediate containers - Intermediate containers shall conform to Specification LLL-B-631, LLL-B-636 or JAN-P-108 Quantities of packing rings with spring or quantities of solid carbon stocks per intermediate container shall be as specified in the contract or order

5.2 Packing -

5.2.1 For domestic shipment, immediate use - Carbon packing rings or solid carbon stocks, packaged as specified in 5.1.1, shall be packed in containers in a manner to insure safe delivery and acceptance at destination Containers shall comply with the Consolidated Freight Classification Rules or other common carrier regulations as applicable to the mode of transportation.

5.2.2 For domestic shipment and storage - Carbon packing rings with springs or solid carbon stock, packaged as specified in 5.1.2, shall be packed in wood-cleated fiberboard, wood-cleated plywood, wirebound, straparound, wood-cleated kraft veneer, or fiberboard boxes conforming to Specification NN-B-591, PPP-B-601 (domestic type), NN-B-631, PPP-B-575, MIL-B-10377 (domestic type) or the special requirements of Specification LLL-B-631 or LLL-B-636 Boxes shall be closed in accordance with the applicable box specification The gross weight of wood or wood-cleated containers shall not exceed approximately 200 pounds gross weight. The gross weight of fiberboard boxes shall not exceed 70 pounds

5.2.3 For overseas shipment - Carbon packing rings with springs or solid carbon stock, packaged as specified in 5.1.2, shall be packed in wood-cleated fiberboard, wood-cleated plywood, wood-cleated kraft veneer board, nailed wood, or fiberboard boxes conforming to Specification JAN-P-103, PPP-B-601 (overseas type), MIL-B-10377 (overseas type), JAN-P-106 or JAN-P-108 Boxes shall be closed and strapped in accordance with the applicable container specification and appendix thereto Wood or wood-cleated boxes shall be lined with a sealed caseliner conforming to Specification MIL-L-10547, except that where sealed intermediate containers conforming to Specification JAN-P-10- are utilized, caseliners will not be required The gross weight of wood or wood-cleated containers shall not exceed 200 pounds, and that of fiberboard boxes shall not exceed 70 pounds

5.3 Marking - In addition to any special marking required by the contract or order, marking for shipment shall be in accordance with Standard MIL-STD-129.

6 NOTES

6.1 Intended use - Carbon stock (solid type) is intended for the manufacture of carbon packing ring

6.2 Ordering data - Procurement documents should specify the following

(a) Title, number, and date of this specification
(b) Class required (see 1.2)
(c) Form and dimensions (see 3.3)

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(d) Condition and finish when the requirements of 3.3 1.5 (c) apply
(e) Whether for domestic shipment, immediate use, domestic shipment and storage, or overseas shipment (see 5.1 and 5 2)
(f) Quantity of carbon stock per containers (see 5.1.2)

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Custodian
Bureau of Ships
Other interest
Navy - Y

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